Sergio A. **Barbosa**

MEng in Computer Science and Systems Engineering | MSc in Artificial Intelligence

Bogotá, Colombia

(+57) 314 283 78 91 | sergioa_barbosas@javeriana.edu.co | sergioabarbosas.github.io/

Profile

I am passionate about software development and artificial intelligence (AI), with a particular interest in the design and implementation of ML models and intelligent systems. I have a background in computer science and experience in developing AI solutions across a range of business and social sectors. My work focuses on designing technological solutions and applying data-driven approaches to solve complex problems.

Education

Pontificia Universidad Javeriana — *Bogotá, Colombia*

MEng in Computer Science and Systems Engineering — *Feb. 2024 - Dec. 2024*

- Degree with honours, *Magna Cum Laude*. Recognition of Academic Excellence (GPA 4.7/5.0).

Pontificia Universidad Javeriana — *Bogotá, Colombia*

MSc in Artificial Intelligence — *Jul. 2021 - Dec. 2022*

- Degree with honours, *Cum Laude*. Recognition of Academic Excellence (GPA 4.6/5.0).

The Naval School 'Almirante Padilla' — *Cartagena, Colombia*

BSc. in Electronic Engineer — *Jan. 2014 - Dec. 2018*

- Degree with honours. Scholarship for excellent academic results (GPA 9.11/10).

Work Experience

Office of the Delegate for Criminal Finances, Prosecutor General's Office — *Bogotá, Colombia*

Research Expert — *Jul. 2025 - Present*

- My primary role is to design and implement technological solutions that support the office's activities. My competencies include software design and development, artificial intelligence, natural language processing, and data science.

Digital GHENOVA (Headquarter Sevilla, Spain) — *Remote from Bogotá, Colombia*

Engineer of AI/NLP — *Jul. 2022 - Jul. 2025*

- Expertly developed innovative technological solutions based on language models for the creation of conversational agents for digital transformation, applied in areas such as healthcare and business. My skills include AI generative techniques, NLP, fine-tuning Large Language Models (LLMs), designing conversational agents, applying DL and ML techniques, OCR applications, dialogue management, and speaker diarisation.

Ministry of Information Technologies and Communications (MINTIC) — *Bogotá, Colombia*

Engineer of AI/NLP (Contractor) — *Oct. 2024 - Dec. 2024*

- Develop and implement AI/NLP algorithms to assess data quality, establishing quality criteria for the open data published on the portal of the MINTIC of Colombia. Skills include working with language large models, deep learning and data science.

(JEP) Special Jurisdiction for Peace Organisation — *Bogotá, Colombia*

Engineer of NLP (Contractor) — *Sep. 2023 - Dec. 2023*

- Successfully adapted and deployed a Named Entity Recognition (NER) model pre-trained in the social context of victims and the armed conflict in Colombia, improving the accuracy and understanding of narratives surrounding this complex socio-political landscape. Skills include working with pre-trained NER models, Transformer Language Models (TLMs), integrated DL models, implementing ML models, and data science.

(COTECMAR) Science and Technology Corporation — *Cartagena, Colombia*

Engineer & Naval Officer — *Jan. 2016 - Dec. 2021*

- Served as a Naval Officer in the Colombian Navy for 16 years, contributing to the production team for the construction of Colombia's first Marine Scientific Research Vessel and other maritime patrol vessels. Developed leadership, problem-solving, resource optimisation, management, and process improvement skills throughout service.

Publications

Detection of Hate Speech, Racism and Misogyny in Digital Social Networks: Colombian Case. (Q1, IF: 5.3) — *MDPI Journal*

Academic publication. Big Data and Cognitive Computing. 8, 113. https://doi.org/10.3390/bdcc8090113 — *Sept. 2024*

- The main contribution of this work is the fine-tuning of a model using a proprietary dataset extracted from social networks, georeferenced in Colombia, and manually labelled with typical and dialect-specific features. Indeed, the experiment compares the performance of TLM models adjusted to the Colombian slang domain against a GPT, having a significant impact on achieving more accurate predictions in this task.

Patente Nacional P202331094. System for Implementing a Digital Navigation Notebook Managed by Voice and Integrated with the Data Acquisition System of a Vessel, and Application Procedure (Spanish Patent and Trademark Office). *Seville, Spain*

Publication No. ES2968547 https://ceo.oepm.es/detalleExpediente?numExp=P202331094 *Jul. 2022 - Oct. 2024.*

- Member of the Inventor Team, implementing AI/NLP models to develop an integrated conversational intelligent system for acquiring data from vessel devices and the operational crew.

Honours & Awards

2025	**Magna Cum Laude**, Degree with honours in Master's in Computer Science.	*Bogotá, Colombia*
2025	**Meritorious thesis in Master's in Computer Science**, Designed and implemented a modular and scalable architecture for a conversational agent focused on querying the Truth Commission's legacy, using a multilingual Retrieval-Augmented Generation (RAG) system with Large Language Models for precise searches in both English and Spanish.	*Bogotá, Colombia*
2023	**Cum Laude**, Degree with honours in Master's in Artificial Intelligence.	*Bogotá, Colombia*
2022	**Meritorious thesis in Master's in Artificial Intelligence**, in computer vision achieved by implementing an AI model utilising Reinforcement Learning algorithms (RL) and Convolutional Neural Network (CNN) to label students' emotions during virtual classes, based on their facial features, gaze, and gestures.	*Bogotá, Colombia*
2022	**Academic Distinctions**, Recognitions of Academic Excellence (3x).	*Bogotá, Colombia*
2018	**Medal "Francisco José de Caldas" category "to consecration"**, Awarded for having obtained the best grade-point average in the academic program of Electronic Engineering.	*Cartagena, Colombia*
2017	**Order of Naval Merit "Almirante Padilla" category "Officer"**, For military virtues, dedication to work, distinguished professional services, exemplary discipline.	*Cartagena, Colombia*

Research Experience

2024 CFP Tech Ethics Lab, University of Notre Dam *Bogotá, Colombia*

Team Member of Project *Jun 2024 — Dec. 2024*

- The goal of this project is to create a Question-Answering System for the Colombian Truth Commission, the first-ever digital archive of the peace process. I am part of the research team at Pontificia Universidad Javeriana (PUJ) and participate in the development of the project.

Session class in Data Analytics Program at Pontificia Universidad Javeriana *Bogotá, Colombia*

Lecturer *Oct. 2023*

- Predictive analytical methods of Machine Learning: decision trees, regression, ensemble models.

Center of Excellence in Big Data & Data Analytics *Bogotá, Colombia*

Develop Team of Project *Jul 2022 — Dec 2022*

- Successfully development of research project about text classification in the social context. Course advanced topics of AI.

Courses & Certificates

2026	**Foundations of Quantum Computing Conference**, dates 2-4 September 2026, University College London (UCL)	*In person*
2025	**SOE-YEEQMSE01: Quantum Mechanics for Scientists and Engineers 1**, Stanford EdX (Online)	*Currently in progress*
2025	**Quantum Computing in Practice**, IBM Quantum Learning (Online)	*Currently in progress*
2025	**Software Design and Architecture Specialisation**, University of Alberta.	*Completed (Online)*
2025	**The Complete Quantum Computing Course**, Udemy	*Completed (Online)*
2024	**Introduction to Quantum Computing**, University College London (UCL). Asynchronous with live practice sessions).	*Completed (Online)*
2023	**6.86x: Machine Learning with Python-From Linear Models to Deep Learning**, MITx EdX.	*Completed (Online)*
2022	**SQL for Data Science**, University of California, Davis.	*Completed (Online)*
2022	**Microsoft Azure Databricks for Data Engineering**, Microsoft.	*Completed (Online)*
2022	**Diploma certificate in Artificial Intelligence**, Sergio Arboleda University.	*Completed (Online)*

Languages

English Professional Working Proficiency. Currently working towards C1 certification
French Upper-Intermediate (B2)